

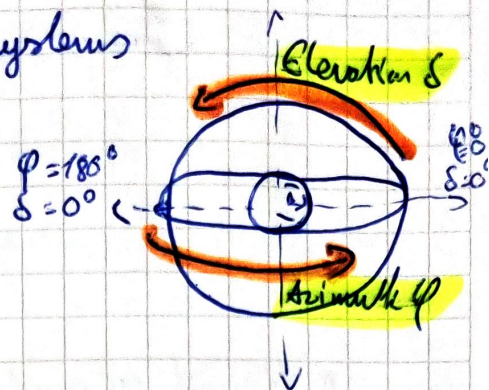
Bio-inspired Robot Sound Localization

- Localisation of sound sources in biological systems

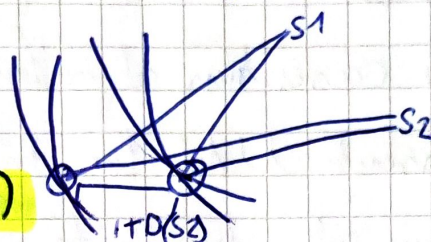
→ Sound has two coordinates

→ azimuth φ

→ elevation δ



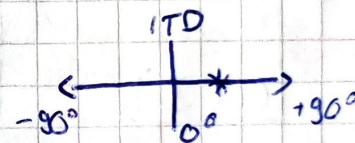
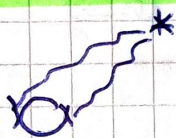
- Difference in azimuth changes time delay of signal arriving at the receivers



$$ITD(S2) > ITD(S1)$$

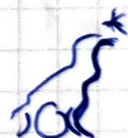
- Interaural Time Difference (ITD)

→ Time difference (ms) between arrival of the signal at the two ears



- Interaural Level Difference (ILD)

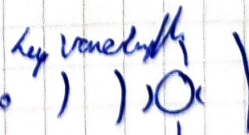
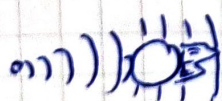
→ Level difference (dB) of the signal at the two ears



- Acoustic Shadow

→ Interaural Level Difference (ILD) best for high frequency sounds ("head shadow effect")

⇒ high difference for high frequencies



⇒ small difference for low frequencies

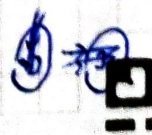


Judgement dominated by

→ ITD in low frequencies

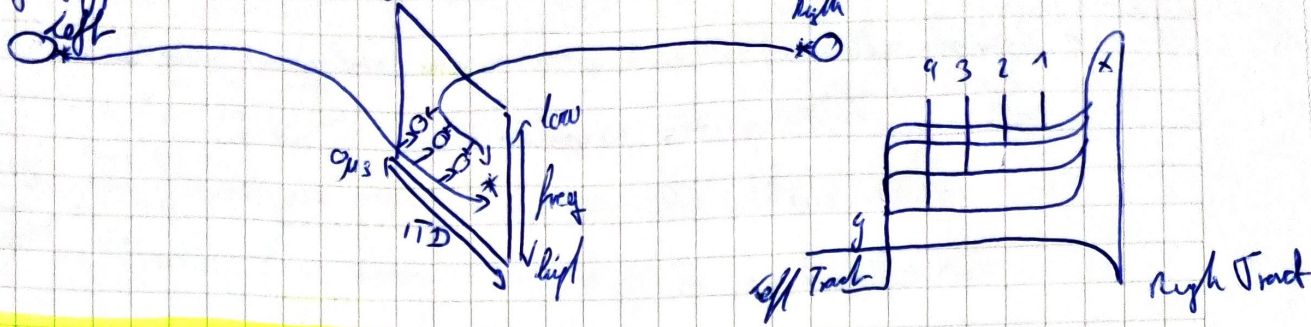
→ ILD in high frequencies

- Difference in elevation also leads to different frequency response



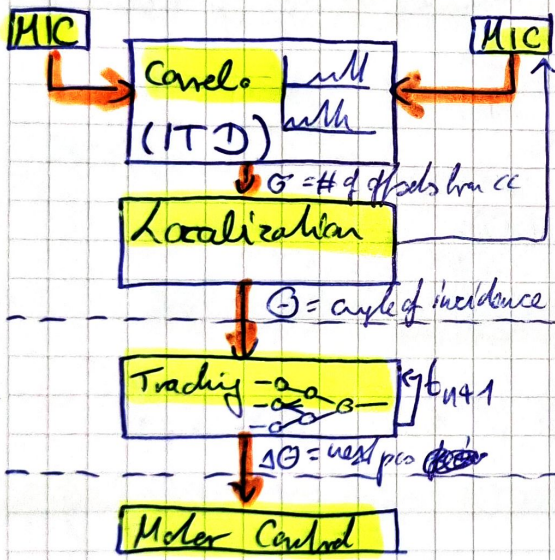
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- Early Jeffers neuronal coincidence model



- Hybrid acoustic tracking

- Inspired by mammalian system (2 ear model, ITD)
- Cross correlation algorithm for sound localization
- Recurrent NN for tracking sound source to improve the localization data
- Learning and adaptation to acceleration and deceleration



Digital representation of detected signals

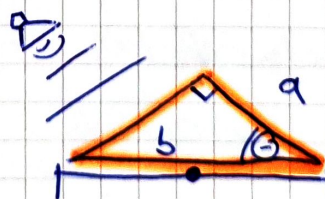
Angle position representing location of source

Next predicted position / angle of source

→ speed of sound

$$a = c * t_{ITD}$$

$$\Theta = \arccos\left(\frac{a}{b}\right)$$

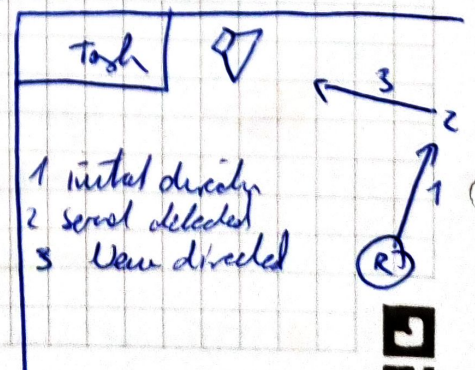


- Geometric Computation (with ITD)

eg: delay $d = 8$ at sample rate of $r = 32000 \text{ } 1/s$
 $c = 340 \text{ } m/s$, $b = 15 \text{ } cm$

$$a = d * \frac{c}{r} = 0,085m$$

$$\Theta = \arccos \frac{a}{b} = 55,48^\circ$$



$$\begin{bmatrix} g(t) \\ h(t) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 2 & 3 & 2 & 1 & 1 \\ 1 & 1 & 1 & 1 & 2 & 3 & 2 & 1 \end{bmatrix}$$

• Cross correlation for localisation

- Determine maximum similarity between two signals $g(t)$ & $h(t)$
- Correlation vector represents ITD delay between signals
- Allows then to determine the angle of incidence of the source

Algo:

- 1) Determine maximum similarity between two signals
- 2) Input: 2 vectors representing digitally sampled sound signals $g(t)$ & $h(t)$

If vectors are not same length, shorter vector is zero-padded to length of longer vector

After padding

Cross-correlation offsets signal $h(t)$ to the left of $g(t)$

Output: calculates sum of product of values of overlapping $g(t)$ and $h(t)$ values

1) Set delay $d = |\vec{g}|$ and similarity $s = 0$

2) For $d_i = |\vec{g}| - 1$ to $-(|\vec{g}| - 1)$

i) zero-pad $\vec{g}$ from right to left and $\vec{h}$ then left to right by d_i zeros

ii) $s_i = \vec{g}^T \vec{p} = \vec{h}^T \vec{p}$

iii) If $s_i \geq s$ then $s = s_i, d = d_i$

3) Return s, d

1 1 2 3 2 1 1 1 $\xrightarrow{d}$
 $\xleftarrow{\quad}$ 1 1 1 1 2 3 2 1

Similarity is the max dot product of shifted 0-padded inputs.

Delay is the time difference at maximal similarity

Bab - inspired

AI

- KV

02.02.2024

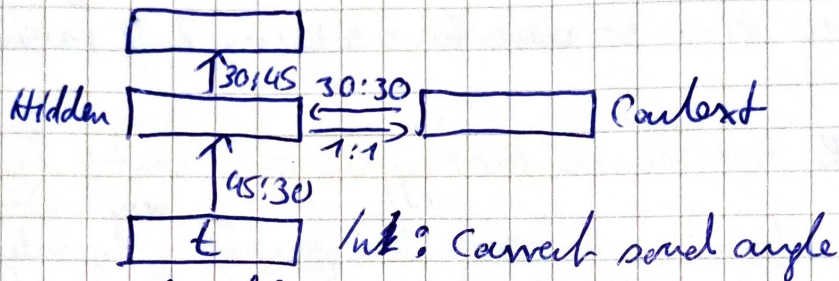


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Simple RNN for prediction of sound angle for moving sound source

SRN as predictor for next angle in trajectory of source

out: predicted sound angle



first input prediction $\rightarrow$ second input $\rightarrow$ now can predict next angle
 t_n t_{n+1} t_{n+2}

$\rightarrow$ two inputs for context needed
 $\rightarrow H_{old} + V_N$

2. Why hybrid localisation and tracking

$\rightarrow$ Algorithmic sound source localisation well understood algorithmically

$\rightarrow$ Cross-correlation does not require training to provide azimuth angle

$\rightarrow$ Neural predicting of source enables a quicker response and can learn temporal sequences

LS

• Cocktail party problem

→ humans & animals have abilities of sound localisation and sound perception in auditory cluttered environment

• Cognitive Neuroscience for sound localisation

→ How is sound encoded?

→ ear pinna → middle ear → inner ear → auditory nerve

→ Tonotopic representation

→ How is the encoded information processed?

→ Spiking NN

→ Interaural level difference (ILD) in Lateral Superior Olive (LSO)

→ Interaural time difference (ITD) in ~~Lateral~~ Medial Superior Olive (MSO)

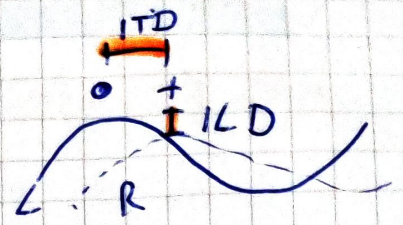
→ Integrated in the Inferior Colliculus (IC)

→ for sound localisation is a major centre of integration in the ascending as well as descending auditory pathways

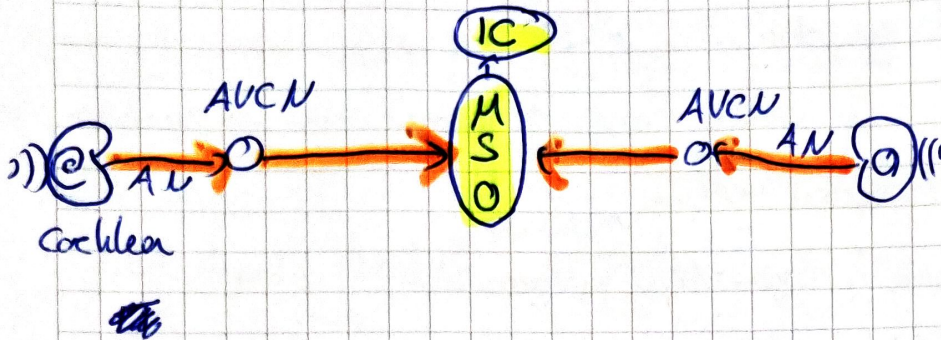
→ Sound encoding: From sounds to spike trains

→ Spikes encoding time and level information

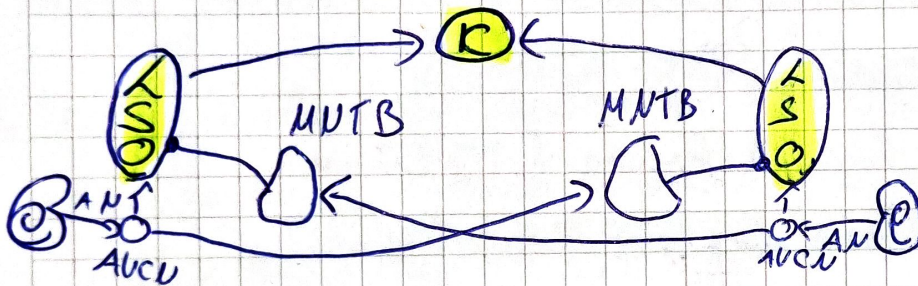
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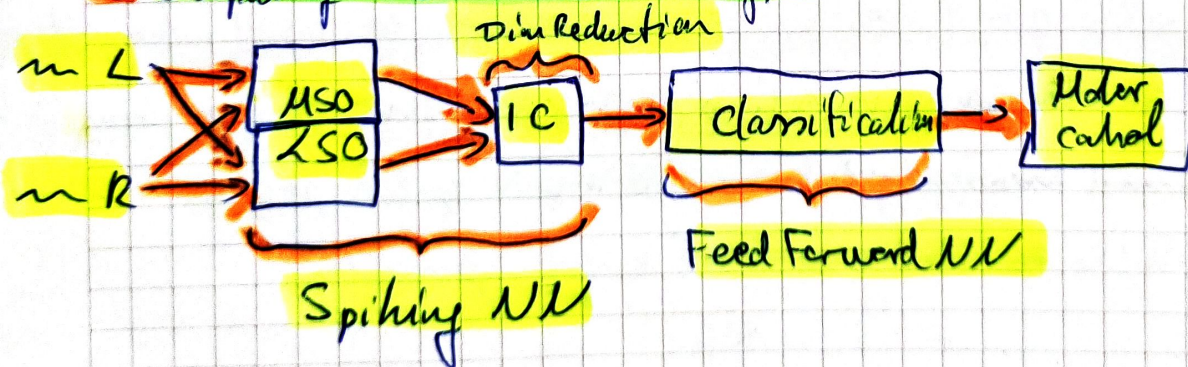
- Bio-inspired sound source localisation
 - Interaural Time Difference (ITD)
 - Interaural Level Difference (ILD)
- ITD at Medial Superior Olive (MSO)



- ILD at Lateral Superior Olive (LSO)



⇒ Output of MSO & LSO integrated in IC



- Summary:

- ITD in a first Hybrid Acoustic System
- ITD & ILD in hybrid Spiking NN
- Approach:
 - New insight into brain mechanism
 - Improve speech recognition by user localisation and vision